

Team Biweekly Report Summary

Project: Smart Maintenance Platform for Aero Engine Industrial Equipment

Period: April 7 - April 20, 2025

Progress Overview

The team has successfully completed the initial design phase tasks, with significant progress on database architecture, AI model design, interface components, and development environment setup. All team members have contributed to their assigned WBS tasks, with several key deliverables completed ahead of schedule. The foundation for both frontend and backend development has been established, allowing for a smooth transition to implementation in the upcoming period.

Project Progress Status

According to the project Gantt chart, during this reporting period (April 7 - April 20, 2025), the team has completed or made significant progress on the following WBS tasks:

- **Task 3.2 (AI Model Design):** 100% complete (Xuanhe Yang)
- **Task 3.4 (Database Design):** 100% complete (Tong Li)
- **Task 3.5 (Core Function Design):** 100% complete (Xuanhe Yang)
- **Task 3.3 (Interface Design):** 70% complete (Jingxiao Han)
- **Task 3.6 (Non-Core Function Design):** 50% complete (Xuanhe Yang)
- **Task 4.1 (Development Environment Setup):** 100% complete (Jingxiao Han)
- **Task 4.3 (Database Implementation):** 80% complete (Tong Li)
- **Task 4.5 (AI Model Training):** 50% complete (Xuanhe Yang)

Current project progress is aligned with the schedule, with several tasks being completed ahead of schedule. The team has successfully transitioned from the design phase to the beginning of implementation, maintaining momentum across all workstreams.

Team Contributions in Relation to WBS

- **Xuanhe Yang (31 hours):**
 - WBS 3.2: AI Model Design (16 hours) - 100% Complete
 - WBS 3.5: Core Function Design (UI+backend) (8 hours) - 100% Complete
 - WBS 3.6: Non-Core Function Design (UI+backend) (4 hours) - 50% Complete
 - WBS 4.5: AI Model Training (3 hours) - 50% Complete
- **Fubin Chen (36 hours):**
 - WBS 3.3: Interface Design (18 hours) - 75% Complete
 - WBS 3.4: Backend Architecture Design (18 hours) - 80% Complete
- **Tong Li (32 hours):**
 - WBS 3.4: Database Design (16 hours) - 100% Complete
 - WBS 4.3: Database Implementation (16 hours) - 80% Complete
- **Jingxiao Han (30 hours):**
 - WBS 2.2: Test Plan Development (12 hours) - 100% Complete
 - WBS 4.1: Development Environment Setup (4 hours) - 100% Complete
 - WBS 3.3: Interface Design (14 hours) - 70% Complete

Key Deliverables

1. **Development Infrastructure:**
 - Complete development environment with frontend, backend, and ML components
 - CI/CD pipelines for automated testing and deployment
 - Development environment documentation

2. **Database Architecture:**
 - ER diagrams with normalized table structures
 - Schema definitions with optimization for time-series data
 - ORM mappings and repository implementations
3. **AI Model Design:**
 - Architecture diagrams for skip-GANomaly and CNN-LSTM models
 - Model parameter specifications and implementation guidelines
 - Initial model training framework with preprocessing pipelines
4. **User Interface Design:**
 - UI mockups for all core system modules
 - Component library and design system
 - Interactive prototypes for usability testing
5. **Backend Architecture:**
 - API specifications for all system modules
 - Service layer architecture patterns
 - Data flow diagrams and authentication framework

Challenges & Solutions

- **Challenge:** Balancing computational requirements for complex ML models with available resources
 - **Solution:** Implemented progressive training strategies and optimized model architectures
- **Challenge:** Designing database schema supporting both transactional operations and time-series analytics
 - **Solution:** Developed hybrid database approach with specialized structures for different data types
- **Challenge:** Integrating diverse technology stacks across frontend, backend, and ML components
 - **Solution:** Created modular architecture with containerization and well-defined APIs
- **Challenge:** Optimizing database performance for high-volume sensor data
 - **Solution:** Implemented partitioning strategies and specialized indexing for time-series data

Next Steps

1. Complete remaining interface designs for report generation and data simulation
2. Finalize backend architecture design elements and begin implementation
3. Continue AI model training and validation
4. Begin core system development with focus on database and API implementations
5. Prepare for system integration testing
6. Develop frontend components based on approved designs

Resource Summary

- **Total Working Hours:** 129 hours
- **Comparison to Plan:** The team has utilized approximately 23.5% of the total project hours (549 hours)
- **Schedule Status:** On track with 1 task ahead of schedule (3.5), 2 tasks at risk of delay (3.3, 3.6)
- **Project Health:** Overall health is good, with minor adjustments needed to ensure Interface Design completion

Biweekly Report

Name: Xuanhe Yang

Period: April 7 - April 20, 2025

Project: Smart Maintenance Platform for Aero Engine Industrial Equipment

Week 3: AI Model Design and Core Function Planning (April 7-13, 2025)

During this week, I focused on designing the AI models that will form the backbone of our predictive maintenance platform. My primary contributions were:

1. **AI Model Design (April 7-13)**
 - Completed detailed architecture design for the skip-GANomaly anomaly detection model
 - Designed CNN-LSTM architecture for remaining useful life (RUL) prediction
 - Completed architecture diagram and parameter specifications for all models

Time spent: 16 hours

Week 4: Advanced Functions and Model Training (April 14-20, 2025)

This week was dedicated to completing the remaining design work and beginning the model training process:

1. **Non-Core Function Design (UI+backend) (April 17-20)**
 - Designed Monitoring Center interface and backend
 - Started Alert System design
2. **AI Model Training (April 14-20)**
 - Prepared MVTec AD dataset for anomaly detection model training
 - Started training skip-GANomaly model on turbine blade images
 - Began preprocessing NASA N-CMAPSS dataset for RUL prediction

Time spent: 15 hours

Completed WBS Items

- **3.2 AI Model Design** (WBS Item, 30 hours) - 100% Complete
- **3.5 Core Function Design (UI+backend)** (WBS Item, 30 hours) - 100% Complete
- **3.6 Non-Core Function Design (UI+backend)** (WBS Item, 24 hours) - 50% Complete
- **4.5 AI Model Training** (WBS Item, 12 hours) - 50% Complete

Challenges & Solutions

The primary challenge this period was balancing the computational requirements of our complex models with available resources. The skip-GANomaly model requires significant GPU resources for training, and initial experiments showed slower convergence than expected. I addressed this by:

1. Implementing a progressive training strategy that gradually increases model complexity
2. Optimizing batch sizes and learning rates based on hardware constraints
3. Refactoring the model to use more efficient implementations of certain layers

Another challenge was integrating the anomaly detection and RUL prediction models into a cohesive system. I developed a modular architecture that allows both models to work independently while sharing relevant information through a common data interface.

Next Steps

1. Complete the remaining Non-Core Function Design (Alert System and Report Generation Module)
2. Continue AI model training and begin validation procedures
3. Start Core Function Development with initial focus on Model Management System
4. Begin preparing the integration framework for connecting AI models with the web application

Total Hours Worked

Total hours for this reporting period: 31 hours

Biweekly Report

Name: Jingxiao Han

Period: April 7 - April 20, 2025

Project: Smart Maintenance Platform for Aero Engine Industrial Equipment

Week 3: Development Environment Setup and Test Planning (April 7-13, 2025)

During this week, I focused on establishing the development infrastructure and creating comprehensive test plans for our predictive maintenance platform. My specific contributions included:

1. **Development Environment Setup (April 4-7)**
 - Configured and deployed the complete development environment for the team
 - Created CI/CD pipelines for automated testing and deployment
 - Documented the environment setup process for team reference
2. **Test Plan Development (April 7-11)**
 - Drafted comprehensive test strategy document
 - Created detailed test plans for all system modules
 - Designed integration testing approach

Time spent: 16 hours

Week 4: Interface Design and Backend Development Preparation (April 14-20, 2025)

This week was dedicated to designing user interfaces and preparing for backend development:

1. **Interface Design (April 17-20)**
 - Designed user interface mockups for core system modules
 - Established UI component library and design system
 - Designed responsive layouts for multi-device support
2. **Backend Service Planning (April 14-16)**
 - Defined API specifications and endpoints
 - Planned database schema and relationships
 - Established backend architecture patterns

Time spent: 14 hours

Completed WBS Items

- **2.2 Test Plan Development** (WBS Item, 12 hours) - 100% Complete
- **4.1 Development Environment Setup** (WBS Item, 4 hours) - 100% Complete
- **3.3 Interface Design** (WBS Item, 24 hours) - 70% Complete

Challenges & Solutions

The main challenge I encountered was establishing a development environment that supports both web application development and machine learning model integration. This required balancing different technology stacks and ensuring compatibility across components. I addressed this by:

1. Creating a modular architecture with clear separation between frontend, backend, and ML components
2. Implementing containerization to isolate dependencies and ensure consistency
3. Establishing well-defined APIs for communication between different system components

Next Steps

1. Complete the remaining interface designs for report generation and data simulation
2. Begin implementation of Non-Core Function Development with Unit Test
3. Collaborate with Xuanhe Yang on integrating ML models with the frontend

Total Hours Worked

Total hours for this reporting period: 30 hours

Biweekly Report

Name: Fubin Chen

Period: April 7 - April 20, 2025

Project: Smart Maintenance Platform for Aero Engine Industrial Equipment

Week 3: Core Function Design - UI Components (April 7-13, 2025)

During this period, I focused on designing the user interface components for our ML-based device lifespan prediction web application. I created wireframes and interactive prototypes for all five core modules:

- Designed the Device Center interface with equipment registration, listing, and detail views
- Created the Monitoring Center dashboard with real-time parameter visualization and status indicators
- Developed Data Simulation module with parameter configuration and simulation control interfaces
- Designed Alert System interface with notification center and alert classification views
- Created Reporting System with customizable report templates and visualization options

For each module, I developed responsive layout designs that optimize the user experience across different devices.

I established a consistent visual language including color schemes, typography, and component styling to ensure interface cohesion throughout the application.

Time spent on UI Component Design: 18 hours

Week 4: Core Function Design - Backend Architecture (April 14-20, 2025)

This week I focused on designing the backend architecture to support our application's core functionalities:

- Designed the database architecture with MongoDB for device metadata and InfluxDB for time-series sensor data
- Developed entity relationship diagrams for key data models including devices, users, models, and alerts
- Created schema definitions with appropriate indexing strategies for optimized query performance

I developed the data models and database schema to efficiently store and retrieve device information, monitoring data, and prediction results. The database design includes optimization for handling time-series data and supporting complex queries for reporting features.

I designed the API endpoints for all five modules, ensuring they follow RESTful principles and provide the necessary data exchange between frontend and backend components. For each endpoint, I defined:

- Request/response formats with detailed JSON schema documentation
- Authentication and authorization requirements using JWT tokens

Time spent on Backend Architecture Design: 18 hours

Completed WBS Items

- **3.4 Core Function Backend Design** (WBS Item, 30 hours) - 80% Complete
- **3.3 Core Function UI Design** (WBS Item, 24 hours) - 85% Complete

Challenges & Solutions

The main challenge I faced was designing a database architecture that could efficiently handle both traditional relational data and high-volume time-series data from sensors. I addressed this by:

1. Implementing a hybrid database approach with MongoDB for metadata and InfluxDB for sensor data
2. Designing appropriate data partitioning strategies based on time ranges and device categories
3. Creating efficient indexing schemes to support common query patterns

Next Steps

1. Finalize the remaining backend architecture design elements
2. Begin implementation of core database models and API endpoints

Total Hours Worked

Total hours for this reporting period: 36 hours

Biweekly Report

Name: Tong Li

Period: April 7 - April 20, 2025

Project: Smart Maintenance Platform for Aero Engine Industrial Equipment

Week 3: Database Design and Architecture (April 7-13, 2025)

During this week, I focused on designing the database architecture for our predictive maintenance platform. My specific contributions included:

1. **Database Conceptual Design (April 7-10)**
 - Completed E-R diagram design, establishing four core entities
 - Identified key relationships between entities with appropriate cardinality
 - Defined primary and foreign key constraints for data integrity
 - Established foundational business rules for data validation
2. **Database Logical Design (April 10-13)**
 - Developed relational model with normalized tables:
 - Optimized schema for both transactional and analytical workloads

Time spent: 16 hours

Week 4: Database Implementation and Testing (April 14-20, 2025)

This week was dedicated to implementing the database design and creating integration points:

1. **Database Implementation (April 14-18)**
 - Completed ORM mapping using Java JPA for all entities
 - Implemented features including:
 - Created migration scripts for schema versioning
2. **Database Testing and Documentation (April 18-20)**
 - Developed test cases for database operations
 - Prepared comprehensive database documentation

Time spent: 16 hours

Completed WBS Items

- **3.1 Database Design** (WBS Item, 20 hours) - 100% Complete
- **4.3 Database Implementation** (WBS Item, 15 hours) - 80% Complete

Challenges & Solutions

The primary challenge was designing a database schema that efficiently supports both transactional operations and analytical queries on time-series data. I addressed this by:

1. Implementing a hybrid approach with normalized tables for transactional data and specialized structures for time-series monitoring data
2. Designing appropriate partitioning strategies for historical monitoring data to optimize query performance
3. Creating a flexible schema that can accommodate future sensor types without structural changes

Next Steps

1. Complete remaining transaction management implementation
2. Develop advanced query functionality for analytics
3. Implement data migration utilities for release management
4. Begin integration with backend API services

Total Hours Worked

Total hours for this reporting period: 32 hours